Credit Risk Scorecard: Model Benchmark & Threshold Sweep

Dataset: Lending Club Accepted Loans (2007–2018 Q4) | **Date:** 2026-09-04 | **Evaluation:** Stratified Cross-Validation

1. Executive Summary

This report summarizes the comparative benchmark across five model architectures (F1–F9) and documents operational decision threshold optimization (F10–F11). All models were evaluated under strict stratified cross-validation with fold-level WoE refitting to eliminate data leakage. Out-Of-Time (OOT) data was strictly quarantined per regulatory credit risk standards.

2. Model Benchmark Summary (CV Results across F1–F11)

Model Architecture	ROC-AUC (Mean ± Std)	Gini (2*AUC - 1)	KS Stat (%)	PR-AUC (Prevalence ~0.20)	F1 Score (t=0.30)	Train Time	Artifact / Path
Baseline Logistic Regression (F7)	0.7088 ± 0.0109	0.4175	33.48%	0.4825	0.5187	0.53s	models/baseline_pipeline.joblib
Random Forest (F8)	0.7020 ± 0.0122	0.4040	35.14%	0.4885	0.5362	1.48s	experiments/runs/..._random_forest/
XGBoost (F8)	0.7049 ± 0.0147	0.4099	34.63%	0.4759	0.5267	2.92s	experiments/runs/..._xgboost/
Default LightGBM (F8)	0.6857 ± 0.0122	0.3714	33.19%	0.4590	0.4952	2.71s	experiments/runs/..._lightgbm/
Optuna-Tuned LightGBM (F9)	**0.7225 (Best)**	**0.4450**	**36.20%**	**0.5140**	**0.5607**	**11.28s**	**experiments/optuna_best_config.yaml**

Key Insight: The Logistic Regression baseline delivers strong performance (ROC-AUC 0.7088, Gini 41.75%) with sub-second training. Optuna tuning of LightGBM achieved a +3.68% ROC-AUC lift to **0.7225**, securing top overall discriminatory power.

3. Decision Threshold Sweep Table (F1–F11 Operating Spectrum)

Simulated on Out-of-Fold validation loans (N=1,200; Defaults=344, Goods=856). Shows operational trade-off between bad-debt capture (recall) and portfolio origination volume (approval rate):

Threshold (t)	Precision	Recall (Bad Capture)	F1 Score	TP	FP	TN	FN	Approval Rate	Approved Bad Rate
0.05	29.8%	97.4%	0.4567	335	788	68	9	6.4%	11.7%
0.10	31.7%	94.5%	0.4748	325	700	156	19	14.6%	10.9%
0.15	34.6%	89.2%	0.4984	307	581	275	37	26.0%	11.9%
0.20	**37.7%**	**84.6%**	**0.5220**	**291**	**480**	**376**	**53**	**35.8%**	**12.4%**
0.25	39.9%	72.7%	0.5155	250	376	480	94	47.8%	16.4%
0.30	**43.4%**	**64.5%**	**0.5187**	**222**	**290**	**566**	**122**	**57.3%**	**17.7%**
0.35	**48.7%**	**54.7%**	**0.5151**	**188**	**198**	**658**	**156**	**67.8%**	**19.2%**
0.40	50.7%	41.6%	0.4569	143	139	717	201	76.5%	21.9%
0.45	52.4%	31.7%	0.3949	109	99	757	235	82.7%	23.7%
0.50	**52.7%**	**23.0%**	**0.3198**	**79**	**71**	**785**	**265**	**87.5%**	**25.2%**
0.55	58.2%	18.6%	0.2819	64	46	810	280	90.8%	25.7%
0.60	65.1%	11.9%	0.2015	41	22	834	303	94.8%	26.6%
0.70	57.1%	2.3%	0.0447	8	6	850	336	98.8%	28.3%
0.80	0.0%	0.0%	0.0000	0	0	856	344	100.0%	28.7%

4. Recommended Operational Cutoff Strategies

Strategy A: Maximum F1 Score (t* = 0.2179)

Operating Metrics: F1: 0.5243 | Precision: 38.6% | Recall: 81.7% | Approval Rate: 39.3%

Business Rationale: Balances bad-debt loss prevention with commercial origination volume. Captures over 81% of all defaults while maintaining strong volume.

Strategy B: Peak KS Separation / Youden's J (t* = 0.3466)

Operating Metrics: KS Divergence: 31.81% | Precision: 48.1% | Recall: 56.1% | Approval Rate: 67.5%

Business Rationale: Maximizes the statistical vertical separation between default and non-default borrower distributions. Expands lending approval to 67.5%.

Strategy C: Policy Target Recall >= 80% (t* = 0.2179)

Operating Metrics: Bad Capture Rate: 81.7% | Precision: 38.6% | Approval Rate: 39.3%

Business Rationale: Mandates that at least 8 out of 10 defaulting applicants are flagged at origination, holding credit loss reserves strictly within compliance.

Strategy D: Asymmetric Business Loss Minimization (t* = 0.1783)

Operating Metrics: Loss Ratio 5:1 (Charge-Off vs Margin) | Min Loss: $744 | Approval Rate: 33.8%

Business Rationale: Because a defaulted loan incurs ~5x higher dollar loss than the interest margin gained on a good loan, the cost-minimizing cutoff shifts downward to t=0.1783.

5. Quality Assurance & Phase 17 Compliance

All modules from F1 through F11 passed automated verification with **100% pass rate** across 13 test suites (97 passed, 0 failures):

- **F1 Data Loader:** Schema validation, fail-loudly on renamed/missing columns, chunked gzip loading.
- **F2 Cleaning & Leakage:** Removal of all 49 leakage/identifier/settlement columns per leakage_audit.md.
- **F3 Chronological Split:** Exact boundary enforcement (Train <=2015, Dev 2016, OOT >=2017).
- **F4 Feature Engineering:** DTI ratios, FICO mid, credit age, log transforms with division-by-zero guards.
- **F5 WoE / IV Pipeline:** Monotonic binning, IV thresholding, frozen transformation on dev/oot.
- **F6 Evaluation Harness:** ROC-AUC, Gini (2*AUC-1), analytically verified known KS proof (KS=0.50).
- **F7 Baseline Scorecard:** Logistic regression on WoE with fold-level refitting and joblib artifact export.
- **F8 Tree Benchmarking:** Unified interface for Random Forest, XGBoost, and LightGBM.
- **F9 Optuna Tuning:** Stratified CV inside objective, SQLite study persistence, best YAML export.
- **F10 SHAP Explainability:** Global beeswarm, mean |SHAP| ranking, and local borrower explanations.
- **F11 Prediction Pipeline:** Decision thresholding, 300-850 scorecard points, and retail credit risk tiers.